UNIVERSITY OF CENTRAL PUNJAB

(Faculty of ITCS)

Operating Systems (OS)

Assignment No 4 **Date: 10 Jan 2026**

Total Marks: 100 **Due Date: 19 Jan 2026**

Assignment Guidelines:

*Assignment is to be **handwritten** and submitted **individually**.*
*There should be a **Title page : Course Name**, **Assignment No.**, **Student Name**, **Reg No.**, **Section**.*
Assignment should be neat, questions attempted in order, and each question should start on new page.
*There will be a **test** based on assignment on the submission day; **submission 50%, test 50%**.*
Late submission will have 25% penalty per day; after three days of deadline submission will not be accepted.

CLO 3 Questions

1. Given six memory partitions of 300 KB, 600 KB, 350 KB, 200 KB, 750 KB, and 125 KB (in order), how would the first-fit, best-fit, and worst-fit algorithms place processes of size 115 KB, 500 KB, 358 KB, 200 KB, and 375 KB (in order).

   Also compare results of these algos; for each placement algo, how many processes allocated, and what small/large fragments were created.

2. Let 15000 be a logical address, in decimal. If page size is 2 KB, in which page does this address exist; also find its offset from the start of page.

3. Consider a computer system with a 128 bit logical address and 4 KB page size. The system supports up to 2 GB of physical memory. How many entries are there in the page table, and how many frames are there.

4. A system has 32 bit L.A and main memory of 64 GB. The page size is 4 KB. Compute the number of pages and frames that exist in the system. Also give the format of L.A and P.A, along with the number of bits needed to represent p, d & f.

5. Consider a system with 64 bit logical address space, and page size of 4 KB. Assuming that each page table entry consists of 4 bytes, how much memory (in bytes) would be required to hold the page table.

6. Consider the following reference string:
   a. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5, 2, 3, 5
   b. 4, 5, 6, 7, 6, 5, 4, 3, 2, 1, 5, 3, 2, 2, 3, 5, 6, 7, 8

   Using FIFO, Optimal and LRU page replacement algorithms, determine the number of page faults for a system with 3 & 4 frames.

7. Short answers:
   a. What are symbolic, relocatable and absolute addresses.
   b. At what stages in a program life, binding of instructions & data to memory addresses can be done.
   c. Define thrashing; how can we prevent/reduce thrashing.
   d. Give short definitions for different types of page replacement algorithms.

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